1. Engineering Problem

1.1 Context and Objective

Smart buildings use environmental sensors to monitor variables such as temperature, humidity, pressure and carbon dioxide (CO_2). Missing sensor measurements reduce the completeness of these time series and may affect subsequent analysis of indoor environmental conditions and building operation.

This project investigates:

How accurately can missing temperature measurements in the Monash Smart Infrastructure Lab be reconstructed using numerical interpolation and machine-learning regression methods, and how does reconstruction performance change with missing-data gap size?

Three methods were compared:

- linear interpolation;
- linear regression; and
- decision tree regression.

Linear interpolation provides a simple numerical baseline using neighbouring temperature observations, while the regression models estimate temperature from other available environmental and time-related variables.

The intended application is retrospective repair of incomplete historical temperature records. The aim is not to identify the most complex model, but to determine which method is most appropriate under different missing-data conditions based on accuracy, data requirements and computational practicality.

The scope is limited to the provided dataset. Interpolation requires valid temperature measurements on both sides of a gap, while the regression models assume that pressure, humidity, CO_2 and timestamp information remain available. Therefore, the study does not represent complete sensor-system failure.

2. Inputs, Outputs and Data

The project uses the Monash Smart Infrastructure Occupancy and Environmental Data supplied for MMA3001. The analysed file contained 4,960 time-stamped observations.

Variable	Unit	Use

Time	Timestamp	Measurement order and time features
Temperature	°C	Reconstruction target
Pressure	mb	Regression predictor
Humidity	RH %	Regression predictor
CO_2	ppm	Regression predictor
LVOC	ppb	Audited; excluded from final model
Formaldehyde	µg/m³	Audited; excluded due to missing data

Pressure, humidity and CO_2 were retained as environmental predictors. Time of day was converted into sine and cosine features to represent its 24-hour periodic nature. LVOC did not provide useful variation, while formaldehyde contained substantial missing data.

For **linear interpolation**, the required inputs are the missing timestamp and valid temperature-time observations immediately before and after the gap. Actual timestamps were used because the measurements were irregularly sampled.

For **linear regression and decision tree regression**, the final predictors were pressure, humidity, CO_2, `hour_sin` and `hour_cos`. Rows without the required predictor information were excluded from model fitting and prediction using complete-case filtering.

The primary output is an estimated temperature in °C. During validation, estimates were compared with known temperatures that had been deliberately withheld. Reconstruction accuracy was reported using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

3. Computational Approach

3.1 Linear Interpolation

Linear interpolation estimates a temperature between two known measurements using:

$$T(t) = T_1 + \frac{t - t_1}{t_2 - t_1}(T_2 - T_1)$$

where T_1 and T_2 are known temperatures measured at times t_1 and t_2.

This method was selected as the baseline because it is simple, interpretable, computationally inexpensive and requires no model training. Its main assumption is that temperature changes approximately linearly between the two surrounding observations. This is most defensible for short gaps.

3.2 Linear Regression

Linear regression predicts temperature as a weighted combination of input variables:

$$\widehat{T} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$$

where $\widehat{T}$ is predicted temperature; $x_1 \ldots x_p$ are the predictors, and $\beta_0 \ldots \beta_p$ are coefficients estimated from the training data.

Linear regression was included as an interpretable machine-learning baseline. Unlike interpolation, it does not require neighbouring temperature observations. However, it assumes that temperature can be represented adequately by a linear combination of the predictors.

3.3 Cyclical Time Features

Using hour of day directly as a linear variable creates an artificial discontinuity between hour 23 and hour despite being only one hour apart. Therefore:

$$hour_{sin} = sin(\frac{2\pi h}{24})$$

$$hour_{cos} = cos(\frac{2\pi h}{24})$$

were created to preserve the cyclical structure of time of day.

3.4 Decision Tree Regression

Decision tree regression was included to test whether nonlinear relationships and interactions between environmental variables could improve reconstruction.

The final tree used the same predictors as linear regression. Model complexity was controlled by `max_depth` and `min_samples_leaf`. Excessively complex trees may overfit the training data, so these parameters were selected using chronological validation.

The final settings were:

- `max_depth = 4`
- `min_samples_leaf = 30`
- `random_state = 42`

The overall workflow was therefore: data audit → artificial withholding → reconstruction → model development → validation → gap-sensitivity analysis → decision-tree optimisation → final chronological evaluation.

4. Validation Method and Results

4.1 Validation Design

Naturally missing temperature measurements have no known true value, so model accuracy cannot be measured directly from them. Instead, valid temperature measurements were deliberately withheld and treated as missing. Their original values were retained as ground truth.

For interpolation, neighbouring temperatures were kept available. For regression, temperature was hidden while the environmental predictors remained available.

Because the dataset is a time series, final model development used chronological rather than random splitting. The first 80% of the 4,960 observations formed the development period and the final 20% (992 observations) formed the later final-test period. The development period was further divided chronologically into approximately 80% training and 20% validation data.

Training data were used to fit the regression models, while validation data were used for formal decision-tree parameter selection. The final-test period was excluded from final model fitting and formal hyperparameter selection.

Earlier exploratory analysis used broader withheld observations to investigate features and candidate methods. Therefore, the final period should be interpreted as a chronological holdout for the **final fitting and formal model-selection stage**, rather than as a completely untouched dataset throughout the entire exploratory project.

4.2 Error Metrics

Two metrics were used:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |T_i - \hat{T}_i|$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (T_i - \hat{T}_i)^2}$$

MAE describes typical absolute reconstruction error, whereas RMSE penalises larger errors more strongly. Both are reported in °C.

4.3 Final Test Results

The final comparison used the same **100 isolated withheld measurements** from the chronologically later test period. Adjacent observations were avoided so that interpolation retained valid neighbouring temperatures.

Method	MAE (°C)	RMSE (°C)

Linear interpolation	**0.117**	**0.256**
Linear regression	0.643	0.845
Decision tree regression	0.811	1.058

Linear interpolation produced substantially lower errors than either regression method. Its MAE of 0.117 °C indicates that reconstructed isolated measurements differed from the true temperatures by approximately 0.12 °C on average.

Linear regression was less accurate but outperformed the decision tree on the final chronological test period. The optimised decision tree produced the highest final-test error despite its greater modelling flexibility.

5. Comparison and Optimisation

5.1 Gap-Size Sensitivity

Interpolation was also evaluated for increasingly long missing gaps.

Consecutive missing observations	Interpolation MAE (°C)
1	0.129
3	0.137
6	0.244
12	0.327

Error increased as gap size increased. This is expected because longer gaps increase the separation between the known temperatures defining the interpolation line, reducing local information about behaviour within the gap.

Because sampling was irregular, these gap sizes represent numbers of missing observations rather than fixed elapsed durations.

5.2 Decision Tree Optimisation

Decision-tree complexity was tuned using the chronological training and validation subsets. The selected parameters, `max_depth = 4` and `min_samples_leaf = 30`, achieved:

- **validation MAE = 0.549 °C**
- **validation RMSE = 0.721 °C**

Restricting tree depth and increasing minimum leaf size reduced model complexity and **was used to limit the risk of overfitting** . However, the decision tree still performed worse than linear regression on the later final-test period, showing that additional model flexibility did not improve generalisation in this dataset.

5.3 Engineering Comparison

The three methods provide different trade-offs.

Linear interpolation achieved the highest accuracy and has very low computational and implementation complexity, but it requires valid temperature observations on both sides of the gap.

Linear regression is less accurate but does not require neighbouring temperature measurements. It is therefore useful as a fallback when pressure, humidity, CO_2 and time information remain available.

Decision tree regression can represent nonlinear relationships but introduced additional model complexity without improving final-test accuracy.

At the scale of this dataset, all three approaches were computationally practical; therefore, reconstruction accuracy, robustness and input availability were more important than runtime differences.

The recommended strategy is consequently:

- use **linear interpolation** for short retrospective gaps with valid temperature measurements on both sides; and
- use **linear regression** when neighbouring temperatures are unavailable but the required environmental predictors remain available.

Decision tree regression is not recommended as the primary reconstruction method for this dataset.

6. Limitations and Lessons Learned

Several limitations affect interpretation of the results:

- First, the data are **irregularly sampled**. A gap containing six records, for example, does not necessarily represent the same elapsed time in every part of the dataset. Future analysis could evaluate performance against actual elapsed gap duration.
- Second, interpolation requires valid **measurements before and after a gap.** It cannot directly reconstruct missing values at the start or end of the record and is unsuitable for true real-time prediction when the future observation does not yet exist.

- Third, regression assumes that other **environmental sensors remain operational**. If pressure, humidity or CO_2 are also missing, the final regression models cannot be applied directly.
- Fourth, the results are **specific to the Monash Smart Infrastructure dataset**. Relationships between variables may differ across rooms, buildings, seasons or sensor systems, so external validation would be required before generalising the numerical error values.
- Finally, some available variables were unsuitable for modelling: LVOC provided little useful variation and formaldehyde contained substantial missing data. Additional reliable variables such as occupancy, HVAC state or external weather may improve future reconstruction.

The **main lesson** from the project was that **greater model complexity does not necessarily produce a better engineering solution**. Linear interpolation was the simplest method yet achieved the best accuracy under suitable conditions. The decision tree was more flexible but generalised less effectively.

A second lesson was that validation design strongly affects engineering credibility. Chronological splitting provided a more realistic final assessment for time-series data than randomly mixing earlier and later measurements.

Overall, the appropriate reconstruction method depends on the failure scenario and available information; **no single method is universally superior**.

7. AI-Use Reflection

ChatGPT was used as a supporting tool for concept explanation, methodology review, debugging, code organisation, documentation and report editing. It also assisted in refactoring key notebook operations into reusable source-code functions and in developing unit tests.

AI made a significant contribution as a support and review tool, but the execution of code, inspection of outputs, model choices and engineering interpretation remained my responsibility. Suggestions were tested rather than accepted automatically.

An important example occurred during validation development. Earlier exploratory work used withheld observations distributed more broadly across the dataset. Review of this approach identified that later observations could influence model development and produce an overly optimistic assessment for time-series data. The workflow was therefore revised to use chronological development, validation and final-test periods for formal model selection and final assessment.

This demonstrated that plausible AI suggestions can still contain methodological weaknesses. AI-generated material was therefore checked by executing the code, comparing results with expected behaviour, reviewing assumptions, checking notebook outputs and running unit tests on the reusable source functions. The final test suite passed all five tests.

AI improved the efficiency of debugging and documentation, but its greatest value was as an iterative review tool rather than a replacement for engineering judgement. The project reinforced the need to understand, test and take responsibility for all AI-assisted code and conclusions.

8. References

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